Swetha Sankar

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UAT keywords: AGN host galaxies; Quasars; Interstellar medium

Summary

Research: Observational black hole–galaxy coevolution; AGN feedback in dust-reddened quasars and radio AGN; shocks, outflows, and warm molecular gas in the multiphase ISM; JWST/NIRSpec IFU and near-infrared spectroscopy

Funding: PI of two STScI Director's Discretionary Research Fund awards ($169,628 total); Maryland Space Grant Consortium Observatory Fellowship

Publications: **3** first author, **12** total, **1** white paper; **250** citations, h-index **8**

Presentations: **3** invited talks, **7** contributed talks, **8** posters

Observing: Co-I of JWST (Cycles 2 and 5) and Roman (Cycle 1) programs; data expertise with JWST/NIRSpec IFU, Gemini/GNIRS, ALMA, and Keck

Students supervised: 2 research students

Collaborations: JWST Early Release Science Program Q3D

Leadership: Co-founder of two departmental/inter-departmental programs (Black (Hole) Coffee Journal Club; PENGUIN Mentoring Program); WFSS conference LOC; JHU G-WiP Summit Organizer

Education

Johns Hopkins University, Ph.D. in Astrophysics 2022 – present
Advisors: Andreea Petric, Timothy Heckman, Nadia Zakamska

University of California, Los Angeles, B.S. in Astrophysics 2018 – 2022
Advisors: Benjamin Zuckerman, Carl Melis, Beth Klein

Observing Programs

Co-Investigator Programs:

James Webb Space Telescope, Cycle 5 GO 10815 — *Following the Shocks: JWST IFU Spectroscopy of NGC 4258's Radio Arcs* 2026

Roman Space Telescope, Cycle 1 GO 10335 — *A Spatially Resolved Look at the Relationships between Star Formation and AGN* 2026

James Webb Space Telescope, Cycle 2 GO 3807 — *Deep Grism Spectroscopy of the Complex Environment around an Extremely Red Quasar within an Ultramassive Host at $z = 3$* 2023

Grants and Awards

Grants:

STScI Director's Discretionary Research Fund, PI ($87,986) — *Tracing the Evolution of Obscured Quasars at Cosmic Noon: A Study of AGN Feedback with Gemini/GNIRS* 2027

STScI Director's Discretionary Research Fund, PI ($81,642) — *Investigating Blue Low-Excitation Radio Galaxies: Unveiling AGN Feedback in Star-Forming Radio-Loud Systems* 2024 – 2025

Fellowships:

Maryland Space Grant Consortium Observatory Fellowship 2026 – 2027

Awards:

NSF Graduate Research Fellowship Program, Honorable Mention 2021, 2023

UCLA Departmental Honors in Physics & Astronomy; UCLA College Honors; UCLA Undergraduate Research Center Student Spotlight 2022

Publications

3 first author, 12 total, 1 white paper. As of October 2026, these works have 250 citations (NASA ADS) with an h-index of 8. Full list: NASA ADS public library.

First Author

3. **Sankar, S.**, Petric, A. O., Pelliccia, D., et al. 2026
When Jets Don't Quench: Near-Infrared H_2 in Star-Forming Low-Excitation Radio Galaxies. arXiv:2606.28257
2. **Sankar, S.**, Zakamska, N. L., Rupke, D. S. N., et al. 2025
JWST Q3D Program: Active Galactic Nucleus Photoionization and Shock Ionization in a Red Quasar at $z = 0.4$. The Astrophysical Journal 991, 227
1. **Sankar, S.**, Melis, C., Klein, B. L., et al. 2021
V488 Per Revisited: No Strong Mid-infrared Emission Features and No Evidence for Stellar/Substellar Companions. The Astrophysical Journal 922, 75

Co-Author

9. Ishikawa, Y., Zakamska, N. L., Chen, Y.-C., et al. (incl. **Sankar, S.**) 2025
VODKA: Complex Molecular Gas Dynamics in a kpc-Separation $z = 2.17$ Dual Quasar with ALMA. arXiv:2502.05327
8. Ishikawa, Y., Zakamska, N. L., Shen, Y., et al. (incl. **Sankar, S.**) 2025
VODKA-JWST: Synchronized Growth of Two Supermassive Black Holes in a Massive Gas Disk? A 3.8 kpc Separation Dual Quasar at Cosmic Noon with the NIRSpec Integral Field Unit. The Astrophysical Journal 982, 22
7. Liu, W., Veilleux, S., **Sankar, S.**, et al. 2025
First Results from the JWST Early Release Science Program Q3D: The Fast Outflow in a Red Quasar at $z = 0.44$. The Astrophysical Journal 980, 31
6. Chen, Y.-C., Ishikawa, Y., Zakamska, N. L., et al. (incl. **Sankar, S.**) 2024
VODKA-JWST: A 3.8 kpc Dual Quasar at Cosmic Noon in a Powerful Starburst Galaxy with JWST/MIRI Integral Field Unit. The Astrophysical Journal 968, 92
5. Bertemes, C., Wylezalek, D., Rupke, D. S. N., et al. (incl. **Sankar, S.**) 2024
JWST ERS Program Q3D: The Pitfalls of Virial Black Hole Mass Constraints Shown in a $z = 3$ Quasar with an Ultramassive Host. Astronomy & Astrophysics, 693, 15
4. Vayner, A., Zakamska, N. L., Ishikawa, Y., **Sankar, S.**, et al. 2024
First Results from the JWST Early Release Science Program Q3D: Powerful Quasar-driven Galactic-scale Outflow at $z = 3$. The Astrophysical Journal 960, 126
3. Vayner, A., Zakamska, N. L., Ishikawa, Y., **Sankar, S.**, et al. 2023
First Results from the JWST Early Release Science Program Q3D: Ionization Cone, Clumpy Star Formation, and Shocks in a $z = 3$ Extremely Red Quasar Host. The Astrophysical Journal 955, 92
2. Rupke, D. S. N., Wylezalek, D., Zakamska, N. L., et al. (incl. **Sankar, S.**) 2023
First Results from the JWST Early Release Science Program Q3D: Benchmark Comparison of Optical and Mid-infrared Tracers of a Dusty, Ionized Red Quasar Wind at $z = 0.435$. The Astrophysical Journal Letters 953, L26
1. Veilleux, S., Liu, W., Vayner, A., et al. (incl. **Sankar, S.**) 2023
First Results from the JWST Early Release Science Program Q3D: The Warm Ionized Gas Outflow in $z \sim 1.6$ Quasar XID 2028 and Its Impact on the Host Galaxy. The Astrophysical Journal 953, 56

White Papers

1. Petric, A., Noirot, G., Alberts, S., et al. (incl. **Sankar, S.**) 2026
Surveying the Universe in 4D: Beating Cosmic Variance with Wide-Field Slitless Spectroscopy from HST, JWST, Euclid, Roman, and Beyond. arXiv:2609.23148

Presentations

* = *invited*

Talks

*University of Michigan, Research Seminar — Ann Arbor, MI; 2026
When Jets Don't Quench: Near-Infrared H_2 in Star-Forming Low-Excitation Radio Galaxies

*Johns Hopkins University, Outflows Club — Baltimore, MD; 2026
When Jets Don't Quench: Near-Infrared H_2 in Star-Forming Low-Excitation Radio Galaxies

*HotSci, JHU/STScI — Baltimore, MD; 2026
When Jets Don't Quench: Near-Infrared H_2 in Star-Forming Low-Excitation Radio Galaxies

Illuminating the Active Universe: Multi-wavelength and Multi-messenger Insights into AGN — ESO Headquarters, DE; 2026
When Jets Don't Quench: Near-Infrared H_2 in Star-Forming Low-Excitation Radio Galaxies

Surveying the Universe in 4D: Wide-Field Slitless Spectroscopy from HST, JWST, Euclid, Roman, and Beyond — STScI, Baltimore, MD; 2026
When Jets Don't Quench: Near-Infrared H_2 in Star-Forming Low-Excitation Radio Galaxies

Supermassive Black Holes and Blue Notes (flash talk) — Montréal, CA; 2026
When Jets Don't Quench: Near-Infrared H_2 in Star-Forming Low-Excitation Radio Galaxies

The Origin and Evolution of Supermassive Black Holes — Sesto, IT; 2024
Ionization in a Red Quasar at $z = 0.4$: Results from the JWST Q3D Program

NRAO/GBO Summer Student Symposium — Green Bank, WV; Aug 2021
Molecular Gas Reservoirs and Their Role in Low-Mass Galaxy Pair Evolution

UCLA Earth, Planetary, and Space Sciences Student Research Symposium — Los Angeles, CA; May 2021
V488 Per Revisited: No Strong Mid-infrared Emission Features and No Evidence for Stellar/Substellar Companions

UCLA REU Symposium — Los Angeles, CA; Aug 2020

Posters

TORUS 2026 — Japan; 2026
When Jets Don't Quench: Near-Infrared H_2 in Star-Forming Low-Excitation Radio Galaxies

18th Potsdam Thinkshop: The Role of Feedback in Galaxy Evolution — Potsdam, DE; 2025
Placing Star-Forming Low-Excitation Radio Galaxies on the Feedback Ladder: Implications for Black Hole–Galaxy Co-Evolution

JHU G-WiP Summit — Baltimore, MD; 2025

IAU Symposium 394 — Cape Town, ZA; 2024
Ionization in a Red Quasar at $z = 0.4$: Results from the JWST Q3D Program

Extreme Galaxies — Reykjavík, IS; 2024
Ionization in a Red Quasar at $z = 0.4$: Results from the JWST Q3D Program

JHU G-WiP Summit — Baltimore, MD; 2024

AAS 243 — New Orleans, LA; 2024
Ionization in a Red Quasar at $z = 0.4$: Results from the JWST Q3D Program

AAS 240 — Pasadena, CA; Jun 2022
ALMA ACA Observations of Molecular Gas in TNT Isolated Pairs

Technical Skills

Data & instruments: JWST/NIRSpec IFU, Gemini/GNIRS, ALMA, Keck/NIRC2, TESS **Analysis:** Q3Dfit, PyQSOFit, PypeIt, CASA, Prospector, emission-line and SED fitting, MCMC **Programming:** Python, C++, Mathematica, JavaScript, LaTeX, Unix **Languages:** English, Tamil, Spanish

Students Supervised

William W. Zhang (Johns Hopkins University, undergraduate) — 2026 – present
Rauhit Krishnan (University of Glasgow) — 2023 – present

Scientific Leadership and Service

Black (Hole) Coffee Journal Club, JHU — Co-founder and Organizer — 2024 – present
PENGUIN Mentoring Program, JHU — Co-founder and Organizer — 2024 – present
WFSS Conference (Surveying the Universe in 4D), STScI — Local Organizing Committee — 2026
JHU G-WiP Summit — Organizer — 2024, 2025
Referee for: Astronomy & Astrophysics Journal

Outreach and Teaching

Outreach:
Maryland Space Grant Observatory — Public Night Instructor; train students and lead public observing — 2026 – present
Women in Physical Sciences, JHU — Member — 2022 – present
Women+ in Physical Sciences, UCLA — Membership Coordinator; Vice President — 2018 – 2022

Teaching:

Teaching Assistant, JHU — Introductory Mechanics, Mechanics Laboratory, Electromagnetism, and E&M Laboratory; received among the highest teaching-assistant ratings in course evaluations	2022 – 2023
Private Tutor — High-school mathematics through calculus	2019 – 2021